Sales & Customer Behaviour Insights – Green Cart Ltd.

Prepared by: Paul Anyira
Course: Data Analytics Project – Week 2
Project Title: Sales & Customer Behaviour Insights – Green Cart Ltd.
Date: June 2026

---

1. Introduction

Green Cart Ltd. is a UK-based e-commerce company specialising in eco-friendly household products. The company requested an analysis of its sales, products, and customer data to support its Q2 performance review and future business decisions.

Three datasets were provided:

- Sales Data (transaction records)
- Product Information
- Customer Information

The objective of this project was to clean and merge the datasets, engineer new features, analyse customer and sales behaviour, and identify business insights that could support marketing, operational, and customer retention strategies.

The report focuses on:

- Revenue performance across product categories and regions
- Customer purchasing behaviour
- Loyalty programme performance
- Discount effectiveness
- Delivery performance and delays

Insert Screenshot 1: Overview of merged dataset (`merged_df.head()`)

	order_id	customer_id	product_id	quantity	unit_price	order_date	delivery_status	payment_method	region_x	dis
0	O966977	C00397	P0022	3.0	39.25	2025-07-06	Delivered	Paypal	Central	
1	O696648	C00236	P0023	5.0	18.92	2025-07-06	Delayed	Credit Card	North	
2	O202644	C00492	P0011	1.0	29.68	2025-07-07	Delivered	Bank Transfer	North	
3	O501803	C00031	P0003	1.0	32.76	2025-07-08	Cancelled	Credit Card	Central	
4	O322242	C00495	P0016	1.0	47.62	2025-07-08	Delayed	Credit Card	West	

2. Data Cleaning Summary

Data cleaning was performed separately on each dataset before merging.

Sales Data

The following actions were completed:

- Standardised delivery_status values (e.g., " DELAYED ", "delayed", "DELAYED" → "Delayed")
- Standardised payment_method values
- Converted order_date to datetime format
- Replaced missing discount values with 0
- Filled missing categorical fields with "Unknown"
- Removed invalid records where necessary
- Validated quantity, unit_price, and discount_applied to ensure no negative values existed

Product Data

The following cleaning steps were completed:

- Standardised category names
- Converted launch_date to datetime format
- Checked for duplicates
- Checked for leading and trailing spaces
- Verified supplier information

Customer Data

The following cleaning steps were completed:

- Standardised loyalty tiers (Gold, Silver, Bronze)
- Corrected inconsistent gender values
- Converted signup_date to datetime format
- Filled missing values where appropriate
- Replaced missing categories with "Unknown"

Missing Values Summary

Column	Action Taken
discount_applied	Filled with 0
loyalty_tier	Filled with Unknown
gender	Filled with Unknown
email	Preserved where missing
region	Filled with Unknown

After cleaning, the datasets were merged using left joins to preserve all sales transactions.

Insert Screenshot 2: Missing values summary (`isnull().sum()`)

```
revenue           0
order_week        0
price_band        0
days_to_order     5
email_domain     26
is_late           0
dtype: int64
```

3. Feature Engineering Summary

Several new variables were created to support deeper analysis.

Feature	Description
revenue	Quantity × Unit Price × (1 − Discount Applied)
order_week	ISO week extracted from order date
price_band	Low (<£15), Medium (£15–£30), High (>£30)
days_to_order	Days between launch date and order date
email_domain	Extracted domain from customer email
is_late	Identifies delayed deliveries
signup_month	Month extracted from signup date
revenue_scaled	Revenue normalised using MinMaxScaler

These features improved the ability to analyse customer behaviour and business performance.

4. Key Findings & Trends

Finding 1: Cleaning Products Drive Revenue

Cleaning products generated the highest revenue (£93,825.89), significantly outperforming all other categories.

Category	Revenue (£)
Cleaning	93,825.89
Storage	47,168.52
Outdoors	40,201.29
Kitchen	33,993.04
Personal Care	24,916.64

Insert Screenshot 3: Top 5 Categories by Revenue Bar Chart

	order_week	region_x	revenue
0	14	Central	68.7600
1	14	East	70.1230
2	14	North	195.1840
3	14	West	53.8365
4	15	Central	16.0650

Finding 2: Gold Customers Generate Most Revenue

Gold loyalty members generated over £137,000 in revenue and were the company's most valuable customer segment.

Loyalty Tier	Revenue (£)
Gold	137,092.17
Silver	52,367.01
Bronze	49,086.04

Insert Screenshot 4: Revenue by Loyalty Tier

	loyalty_tier	signup_month	customers	orders	revenue
0	Bronze	2024-07	10	57	4038.7480
1	Bronze	2024-08	10	46	3831.0110
2	Bronze	2024-09	7	36	2685.4130
3	Bronze	2024-10	12	53	4247.2735
4	Bronze	2024-11	9	50	4149.2955

Finding 3: Delivery Delays Are Highest in the East Region

The East region recorded the highest delivery delay rate (41.7%).

Region	Delay Rate (%)
East	41.7
North	39.3
Central	39.0
South	38.6

Region	Delay Rate (%)
West	37.0

Insert Screenshot 5: Delivery Delay Rate by Region

	region_x	price_band	total_orders	delayed_orders	delay_rate
0	Central	Low	113	43	38.053097
1	Central	Medium	225	88	39.111111
2	Central	High	265	104	39.245283
3	East	Low	96	40	41.666667
4	East	Medium	239	102	42.677824
5	East	High	267	109	40.823970
6	North	Low	107	41	38.317757
7	North	Medium	206	91	44.174757
8	North	High	293	106	36.177474
9	South	Low	98	33	33.673469
10	South	Medium	202	73	36.138614
11	South	High	296	124	41.891892
12	West	Low	77	30	38.961039

5. Business Question Answers

Question 1: Which product categories drive the most revenue, and in which regions?

Cleaning products generated the highest overall revenue (£93,825.89).

The strongest regional performance was:

Category	Best Performing Region	Revenue (£)
Cleaning	East	19,975.33
Storage	West	10,014.45
Outdoors	South	9,327.42
Kitchen	South	7,761.40
Personal Care	Central	5,640.52

Cleaning products should remain a strategic focus for future growth.

Question 2: Do discounts lead to more items sold?

The analysis found no meaningful relationship between discounts and quantity sold.

Correlation between discount and quantity:

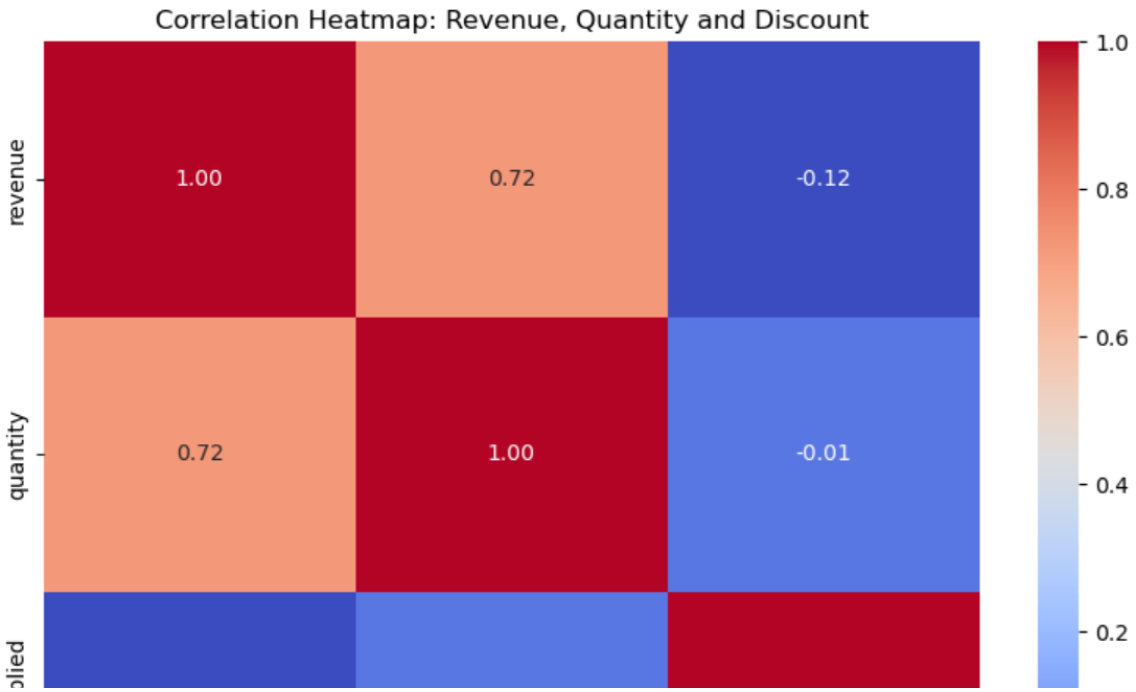
- Correlation = -0.008

Average quantity sold:

Discount Band	Average Quantity
0–10%	3.01
10–20%	2.98

This suggests discounts had minimal impact on customer purchasing volume.

Insert Screenshot 6: Correlation Heatmap



Question 3: Which loyalty tier generates the most value?

Gold customers generated the greatest value.

Loyalty Tier	Revenue (£)	Orders
Gold	137,092.17	1,679
Silver	52,367.01	660
Bronze	49,086.04	625

The loyalty programme appears highly effective at driving customer value.

Question 4: Are certain regions struggling with delivery delays?

Yes.

The East region recorded the highest delivery delay rate (41.7%).

Although delays occurred across all regions, operational improvements should prioritise the East region.

Question 5: Do customer signup patterns influence purchasing activity?

Yes.

The highest-performing signup cohorts were:

Signup Month	Revenue (£)
Aug 2024	24,008.72
Feb 2025	23,829.38
Oct 2024	22,479.31
Nov 2024	22,427.76

These results suggest acquisition timing and marketing campaigns may influence long-term customer value.

Insert Screenshot 7: Revenue by Signup Month Chart

	signup_month	customers	orders	revenue
1	2024-08	49	277	24008.7165
7	2025-02	43	277	23829.3800
3	2024-10	49	275	22479.3110
4	2024-11	45	276	22427.7560
5	2024-12	45	269	20316.3805
8	2025-03	37	241	19886.9960
2	2024-09	42	239	18603.8720
6	2025-01	40	237	17668.7545
0	2024-07	36	221	17618.1430
11	2025-06	34	210	17038.6615
10	2025-05	35	205	15958.9135
9	2025-04	34	197	15268.0680
13	NaT	8	54	3612.9235
12	2025-07	2	21	1008.1560

6. Recommendations

Recommendation 1: Increase Investment in Cleaning Products

Cleaning products consistently outperform all other categories. Future promotional campaigns should prioritise this category, especially within the East region.

Recommendation 2: Expand Loyalty Programme Benefits

Gold customers generate the majority of revenue. Introducing incentives to encourage Bronze and Silver customers to move into the Gold tier could increase customer lifetime value.

Recommendation 3: Improve Delivery Performance in the East Region

The East region experiences the highest delay rate. Reviewing logistics providers, warehouse operations, and fulfilment processes could improve delivery reliability and customer satisfaction.

---

7. Data Issues and Risks

A key data quality issue identified during the project was inconsistent formatting across several fields, including:

- loyalty_tier
- delivery_status
- gender
- date formats

Missing customer information was also identified.

Recommended Improvements

- Implement dropdown menus for categorical fields
- Apply validation rules during data entry
- Introduce automated data quality checks
- Standardise date formats across systems

These improvements would reduce data cleaning effort and improve reporting accuracy.

Stretch Tasks Completed

✓ Customers who signed up in Q2 identified using `.query()`

✓ Orders placed within 14 days identified

✓ Discounts greater than 20% checked

✓ Revenue normalised using MinMaxScaler

✓ Five underperforming products flagged based on:

- Low sales quantity
- High discount levels
- Delayed deliveries

Conclusion

The analysis revealed that Cleaning products are Green Cart Ltd.'s strongest revenue driver, Gold loyalty members generate the greatest customer value, and delivery performance requires improvement in the East region. Customer signup timing also appears to influence future purchasing behaviour. By focusing on high-performing product categories, strengthening loyalty programmes, and improving delivery operations, Green Cart Ltd. can continue to improve revenue growth and customer satisfaction.